

ENGR 200 Fall 2019

Solutions of Midterm 1 Study Questions

1.

(a)

$$P(A|B) > P(A) \Rightarrow \frac{P(A \cap B)}{P(B)} > P(A) \Rightarrow \frac{P(A \cap B)}{P(A)} > P(B) \Rightarrow P(B|A) > P(B)$$

(b)

i. $P(E) = 0.4, P(E|F) = P(E) \Rightarrow P(E|F) = 0.4$

ii. $P(E \cap F) = 0 \Rightarrow P(E|F) = 0$

iii. $F \subset E \Rightarrow P(E \cap F) = P(F) \Rightarrow P(E|F) = \frac{P(F)}{P(F)} = 1$

iv. $E \subset F \Rightarrow P(E \cap F) = P(E) \Rightarrow P(E|F) = \frac{P(E)}{P(F)} = \frac{0.4}{P(F)}$

2.

- (a) We need to find probabilities for $x = 0, 1, 2, 3$. The sample space consists of the $\binom{20}{3}$ different ways that we can select 3 items out of the available items. For the events of the interest to occur, we have to select x defected items out of the 4 defected items, which can be done in $\binom{4}{x}$ ways, and also select $3 - x$ non-defected out of the 16 items that are not defected, which can be done in $\binom{16}{3-x}$ ways. Therefore, we can write probabilities of selecting x defected items as follows:

$$p_X(0) = \frac{\binom{4}{0} \binom{16}{3}}{\binom{20}{3}} = \frac{540}{1140}$$

$$p_X(1) = \frac{\binom{4}{1} \binom{16}{2}}{\binom{20}{3}} = \frac{480}{1140}$$

$$p_X(2) = \frac{\binom{4}{2} \binom{16}{1}}{\binom{20}{3}} = \frac{96}{1140}$$

$$p_X(3) = \frac{\binom{4}{3} \binom{16}{0}}{\binom{20}{3}} = \frac{4}{1140}$$

Then, the probability mass function is as follows:

$$p_X(x) = \begin{cases} \frac{540}{1140}, & x = 0 \\ \frac{480}{1140}, & x = 1 \\ \frac{96}{1140}, & x = 2 \\ \frac{4}{1140}, & x = 3 \\ 0, & o.w \end{cases}$$

(b) Expected value of X is

$$\begin{aligned} E[X] &= 0.p_X(0) + 1.p_X(1) + 2.p_X(2) + 3.p_X(3) \\ &= 0.\frac{540}{1140} + 1.\frac{480}{1140} + 2.\frac{96}{1140} + 3.\frac{4}{1140} \\ &= \frac{684}{1140} \\ &= \frac{3}{5} \end{aligned}$$

(c) To find the variance of X , let us first calculate $E[X^2]$:

$$\begin{aligned} E[X^2] &= 0^2.p_X(0) + 1^2.p_X(1) + 2^2.p_X(2) + 3^2.p_X(3) \\ &= 0.\frac{540}{1140} + 1.\frac{480}{1140} + 4.\frac{96}{1140} + 9.\frac{4}{1140} \\ &= \frac{900}{1140} \\ &= \frac{15}{19} \end{aligned}$$

Then,

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= \frac{15}{19} - \left(\frac{3}{5}\right)^2 \\ &= \frac{204}{475} \end{aligned}$$

3.

(a) We define C and A as

C: batch C is selected

A:one of the two components is defective

$$P(A|C) = \frac{\binom{2}{1}\binom{8}{1}}{\binom{10}{2}} = \frac{16}{45}$$

(b) M:exactly one of the two components is defective

$$P(M) = P(M|A)P(A) + P(M|B)P(B) + P(M|C)P(C) = 0 + \frac{\binom{1}{1}\binom{9}{1}}{\binom{10}{2}} \times 0.3 + \frac{\binom{2}{1}\binom{8}{1}}{\binom{10}{2}} \times 0.2$$

(c) M:both components defective

C:batch C is selected

$$P(C|M) = \frac{P(C)P(M|C)}{P(M)} = \frac{P(M|C)P(C)}{P(M|A)P(A) + P(M|B)P(B) + P(M|C)P(C)} = 1$$

(d) M:none of the two components defective

$$\begin{aligned} P(A|M) &= \frac{P(M|A)P(A)}{P(M|A)P(A) + P(M|B)P(B) + P(M|C)P(C)} = \\ &= \frac{\frac{\binom{10}{2} \times 0.5}{\binom{10}{2}}}{\frac{\binom{10}{2} \times 0.5}{\binom{10}{2}} + \frac{\binom{1}{0}\binom{9}{2} \times 0.3}{\binom{10}{2}} + \frac{\binom{2}{0}\binom{8}{2} \times 0.2}{\binom{10}{2}}} \end{aligned}$$

(e) X:number of defective components,0,1,2

$$E(X) = 0 \times P(X = 0) + 1 \times P(X = 1) + 2 \times P(X = 2)$$

$$\begin{aligned}
&= P(X = 1|A)P(A) + P(X = 1|B)P(B) + P(X = 1|C)P(C) + \\
&2 \times [P(X = 2|A)P(A) + P(X = 2|B)P(B) + P(X = 2|C)P(C)] = \\
&= \frac{\binom{1}{1}\binom{9}{1} \times 0.3}{\binom{10}{2}} + \frac{\binom{1}{1}\binom{8}{1} \times 0.2}{\binom{10}{2}} + 2 \times \frac{\binom{2}{2}\binom{8}{0} \times 0.2}{\binom{10}{2}}
\end{aligned}$$

4. Let X be number of tosses required until you get a sum of 5 or a sum of 7.

(a) First alternative,

$$\begin{aligned}
E[X] &= E[X|\text{a sum of 5 or 7 is achieved}]P(\text{a sum of 5 or 7 is achieved}) \\
&\quad + E[X|\text{a sum of 5 or 7 not achieved}]P(\text{a sum of 5 or 7 not achieved})
\end{aligned}$$

$$\begin{aligned}
E[X] &= 1 \cdot \frac{10}{36} + (1 + E[X]) \frac{26}{36} \\
E[X] &= \frac{36}{10} \\
&= 3.6
\end{aligned}$$

Another alternative is that since the number of tosses required until the end of the game is a geometric random variable with $p = \frac{10}{36}$, then $E[X] = \frac{1}{p} = \frac{36}{10}$.

(b) We are asked to find the probability of achieving a sum of 7 before achieving a sum of 5. Let Y be achieving a sum of 7 before achieving a sum of 5, then:

$$\begin{aligned}
P(Y) &= P(Y|\text{a sum of 7 on first roll})P(\text{a sum of 7 on first roll}) \\
&\quad + P(Y|\text{neither a sum of 7 nor 5 on first roll})P(\text{neither a sum of 7 nor 5 on first roll}) \\
&\quad + P(Y|\text{a sum of 5 on first roll})P(\text{a sum of 5 on first roll}) \\
P(Y) &= \frac{6}{36} + P(Y) \cdot \frac{26}{36} + 0 \cdot \frac{4}{36} \\
P(Y) &= \frac{3}{5}
\end{aligned}$$

(c)

$$\begin{aligned} E[X] &= 3.P(\text{a sum of 5 before a 7}) - 2.P(\text{a sum of 7 before a 5}) \\ &= 3.\frac{2}{5} - 2.\frac{3}{5} \\ &= 0 \end{aligned}$$

(d)

$$\begin{aligned} E[X^2] &= 3^2.P(\text{a sum of 5 before a 7}) + (-2)^2.P(\text{a sum of 7 before a 5}) \\ &= 9.\frac{2}{5} + 4.\frac{3}{5} \\ &= 6 \end{aligned}$$

$$\begin{aligned} Var(X) &= E[X^2] - E[X]^2 \\ &= 6 \end{aligned}$$

5. Let X_i denote the outcome of the i th roll.

(a)

$$\begin{aligned} &P(X_1 < X_4, X_2 < X_4, X_3 < X_4 | X_4 = 3) \\ &= P(X_1 < X_4 | X_4 = 3).P(X_2 < X_4 | X_4 = 3).P(X_3 < X_4 | X_4 = 3) \\ &= \left(\frac{2}{6}\right)^3 \end{aligned}$$

Note that the first equality comes from the independence.

(b) The game is won only when $\max\{X_1, X_2, X_3\} < X_4$

$$\begin{aligned} P(\max\{X_1, X_2, X_3\} < X_4) &= \sum_{j=1}^6 P(\max\{X_1, X_2, X_3\} < X_4 | X_4 = j) \cdot P(X_4 = j) \\ &= 0 \cdot \frac{1}{6} + \left(\frac{1}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{2}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{3}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{4}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} \\ &= \frac{225}{1296} \end{aligned}$$

Note that each die being less than the fourth one is independent of others being less than the fourth one.

(c) Let Y be the payoff of the game. Then:

$$\begin{aligned} E[Y] &= 100 \cdot P(X_4 > \max\{X_1, X_2, X_3\}) - 10 \cdot P(X_4 \leq \max\{X_1, X_2, X_3\}) \\ &= 100 \cdot \left(\frac{225}{1296}\right) - 10 \cdot \left(1 - \left(\frac{225}{1296}\right)\right) \\ &= 9.097 \end{aligned}$$

6.

(a) Let random variable X be the number of inspections required until the flaw is detected.

The probability of a detection of the flaw at the second inspection can be represented as $P(X = 2)$. As the inspections are successive, a detection of the flaw at the second inspection indicates a fail in the first attempt. Then, we need to find the probability of detecting after one not being able to detect. Note that inspections are independent from each other. So,

$$P(X = 2) = (1 - p)p$$

(b) Let D be the event of detecting a flaw in an inspection with probability $P(D) = p$ and D^c be not detecting it with $P(D^c) = 1 - p$. Let random variable X be the number of inspections required until the flaw is detected.

The detection at the n th inspection entails $n - 1$ unsuccessful trial. As the probability of detecting is same for each trial and independent from previous ones, we can write

$$P(X = n) = P(D^c)^{n-1}P(D) = (1 - p)^{n-1}p$$

which is a geometric random variable.

(c) A flawed item passing all three inspections means 3 consecutive failed inspections. Let this event be A . Probability of passing each inspection is $(1 - p)$ and independent from each other, so $P(A) = (1 - p)^3$

(d)

F : chosen ball has a flaw with $P(F) = 0.1$ (given)

F^c : chosen ball has not a flaw with $P(F^c) = 0.9$ (given)

A : randomly chosen item passing 3 inspections

$P(A|F^c) = 1$ since if there is no flaw item definitely passes the inspection

$P(A|F)$: what we calculated in part c.

$$\begin{aligned} P(A) &= P(F)P(A|F) + P(F^c)P(A|F^c) \\ &= 0.1(1 - p)^3 + 0.9 \end{aligned}$$

(e)

$$\begin{aligned} P(F|A) &= \frac{P(A|F)P(F)}{P(A)} \\ &= \frac{0.1(1 - 0.2)^3}{0.1(1 - 0.2)^3 + 0.9} = 0.054 \end{aligned}$$

7.

(a)

$$P(A) = 0.2, P(B) = 0.3, P(C) = P(A \cup B), P(D) = ((A \cap B^c) \cup (B \cap A^c))$$

$$P(C) = P(A) + P(B) - P(A \cap B) = 0.2 + 0.3 - 0.2 \times 0.3 = 0.44$$

(b)

$$P(D) = ((A \cap B^c) \cup (B \cap A^c)) = P(A) - P(A \cap B) + P(B) - P(A \cap B) = P(A) + P(B) - 2P(A \cap B) = 0.2 + 0.3 - 2 \times 0.3 \times 0.2 = 0.38$$

(c)

$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{P(A \cap B^c)}{P(D)} = \frac{P(A) - P(A \cap B)}{P(D)} = \frac{0.14}{0.38} = \frac{14}{38}$$

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.14}{0.2} = \frac{14}{20}$$

(d)

$P(D|A) \neq P(D)$, so D and A are dependent

8.

(a) Number of ways to choose 3 girls out of 12 and 2 boys out of 10.

$$\binom{12}{3} \binom{10}{2}$$

(b) Sample space Ω : choosing 5 people from the whole class so $|\Omega| = \binom{22}{5}$

Event A : Choosing 3 girls and 2 boys (from part (a)) so $|A| = \binom{12}{3} \binom{10}{2}$

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\binom{12}{3} \binom{10}{2}}{\binom{22}{5}}$$

(c) Total class number: $n = 10+12 = 22$

n people can sit around a table in $(n-1)!$ different ways. So answer is $(22-1)! = 21!$

9.

(a) To have a proper pmf, $\sum_{x \in \{1,4,9\}} p_X(x) = 1$ should be satisfied:

$$p_X(1) + p_X(4) + p_X(9) = 1$$

$$a + a\sqrt{4} + a\sqrt{9} = 1$$

$$6a = 1$$

$$a = \frac{1}{6}$$

(b)

$$P(X = 4|X < 9) = \frac{P(X = 4 \text{ and } X < 9)}{P(X < 9)}$$

$$P(X = 4|X < 9) = \frac{P(X = 4)}{P(X < 9)}$$

$$P(X < 9) = P(X = 1) + P(X = 4) = a + a\sqrt{4} = 3a = 3 \cdot \frac{1}{6} = \frac{1}{2}$$

$$P(X = 4) = 2a = 2 \cdot \frac{1}{6} = \frac{1}{3}$$

Then,

$$P(X = 4|X < 9) = \frac{P(X = 4)}{P(X < 9)} = \frac{1/3}{1/2} = \frac{2}{3}$$

(c)

$$E(X) = \sum_{x \in \{1,4,9\}} xp_X(x)$$

$$E(X) = 1 \cdot p_X(1) + 4 \cdot p_X(4) + 9 \cdot p_X(9)$$

$$E(X) = 1 \cdot a + 4 \cdot a\sqrt{4} + 9 \cdot a\sqrt{9}$$

$$E(X) = a + 4 \cdot 2a + 9 \cdot 3a$$

$$E(X) = 36a = 36 \cdot \frac{1}{6}$$

$$E(X) = 6$$

(d)

$$E(X) = \sum_{x \in \{1,4,9\}} \sqrt{x}p_X(x)$$

$$E(X) = 1 \cdot p_X(1) + \sqrt{4} \cdot p_X(4) + \sqrt{9} \cdot p_X(9)$$

$$E(X) = 1 \cdot a + 2 \cdot a\sqrt{4} + 3 \cdot a\sqrt{9}$$

$$E(X) = a + 2 \cdot 2a + 3 \cdot 3a$$

$$E(X) = 14a = 14 \cdot \frac{1}{6}$$

$$E(X) = \frac{7}{3}$$

10.

(a) Let X be the random variable representing the demand for the bicycle. We need to calculate $P(X \geq 1)$:

$$P(X \geq 1) = 1 - P(X = 0)$$

Since X has Poisson distribution with parameter $\lambda = 1$,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-1}1^x}{x!} \text{ for } x = 0, 1, 2, \dots \text{ Hence,}$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-\lambda}\lambda^0}{0!}$$

$$P(X \geq 1) = 1 - e^{-\lambda}$$

$$P(X \geq 1) = 1 - e^{-1} = 0.6321.$$

(b) Let R be the random variable that represents the revenue. When the demand is 0, you are not able to sell your bike, which means that $R = 0$. When the demand is above 1, you are able to sell your bike and the revenue is $R = 100$ TL. Therefore,

expected revenue is:

$$E(R) = 0 \cdot P(X = 0) + \sum_{x=1}^{\infty} 100 \cdot P(X = x) = 100 \sum_{x=1}^{\infty} P(X = x)$$

$$E(R) = 100(1 - P(X = 0))$$

$$E(R) = 100(1 - e^{-\lambda})$$

$$E(R) = 100(1 - e^{-1}) = 63.21 \text{ TL}$$

(c) Since the expected revenue is 63.21 TL, which is greater than 50 TL, we should not sell it at 50 TL.

11.

(a) X has a geometric distribution with the parameter $p = 0.6$:

$$P(X = x) = \sum_{k=1}^x (1-p)^{k-1} p \text{ for } x = 1, 2, \dots$$

$$P(X \leq 2) = p + (1-p)p = 0.6 + (0.4)(0.6) = 0.84$$

(b) For the geometric random variable, $E[X] = 1/p$. Thus, $E[X] = 1/0.6 = 5/3$ minutes.

(c) $C(x) > 4 \Rightarrow x^2/3 + 1 > 4 \Rightarrow x > 3$ $\left(p + (1-p)p + (1-p)^2 p \right)$

$$P(C(X) > 4) = P(X > 3) = 1 - P(X \leq 3) = 1 - (0.6 + 0.24 + 0.16 \cdot 0.6)$$

$$= 1 - (0.6 + 0.24 + 0.16 \cdot 0.6) = 1 - 0.984 = 0.016$$

(d) $E[C(X)] = E[X^2/3 + 1] = E[X^2]/3 + 1$

We know that $E[X^2] = V[X] + E^2[X]$. $V(X) = (1-p)/p^2$ and $E[X] = 1/p$ for the geometric random variable. Thus,

$$E[X^2] = V[X] + E^2[X] = (1-0.6)/0.6^2 + (1/0.6)^2 = 10/9 + 25/9 = 35/9.$$

$$E[C(X)] = E[X^2/3 + 1] = E[X^2]/3 + 1 = (35/9)(1/3) + 1 = 35/27 + 1 = 62/27$$

12.

(a)

$$\begin{aligned}\sum_{x=1}^3 \sum_{y=1}^2 \frac{c}{xy} &= 1 \\ c + \frac{c}{2} + \frac{c}{2} + \frac{c}{4} + \frac{c}{3} + \frac{c}{6} &= 1 \\ 11c/4 &= 1 \\ c &= 4/11\end{aligned}$$

(b) $p_X(x) = \sum_{y=1}^2 \frac{4/11}{xy} = \frac{4/11}{x} + \frac{4/11}{2x} = \frac{4}{11x} + \frac{2}{11x} = \frac{6}{11x}$ for $x = 1, 2, 3$

$$E[X] = \sum_{x=1}^3 x \frac{6}{11x} = \sum_{x=1}^3 \frac{6}{11} = 18/11$$

(c) The minimum value of Z is 1 and the maximum value is 3.

$$P(Z = 1) = p_{X,Y}(1, 1) = 4/11$$

$$P(Z = 2) = p_{X,Y}(1, 2) + p_{X,Y}(2, 1) + p_{X,Y}(2, 2)$$

$$P(Z = 2) = \frac{4/11}{2} + \frac{4/11}{2} + \frac{4/11}{4} = 2/11 + 2/11 + 1/11 = 5/11$$

$$P(Z = 3) = p_{X,Y}(3, 1) + p_{X,Y}(3, 2) = \frac{4/11}{3} + \frac{4/11}{6} = 4/33 + 2/33 = 6/33 = 2/11$$

Then the pmf of Z is

$$p_Z(z) = \begin{cases} 4/11, & \text{if } z = 1 \\ 5/11, & \text{if } z = 2 \\ 2/11, & \text{if } z = 3 \\ 0 & \text{otherwise} \end{cases}$$

(d) $E[Z] = \sum_{z=1}^3 zp_Z(z) = (1)(4/11) + (2)(5/11) + (3)(2/11) = 20/11$

(e) $E[Z^2] = \sum_{z=1}^3 z^2 p_Z(z) = (1)(4/11) + (4)(5/11) + (9)(2/11) = 42/11$

$$V(Z) = E[Z^2] - E^2[Z] = 42/11 - (20/11)^2 = 0.5124$$

13.

(a) $E[X] = 2.a + 0.b + 0.05.(-1) + 0.15.3 + 0.25.(-2) = 2.a - 0.1 = 0.5 \rightarrow a = 0.3$

$$\sum_x p_X(x) = a + b + 0.05 + 0.15 + 0.25 = 0.75 + b = 1 \rightarrow b = 0.25$$

(b) $E[X^2 + 2] = E[X^2] + E[2] = E[X^2] + 2$

$$E[X^2] = \sum_x p_X(x) X^2 = 0.3(2)^2 + 0.25(0)^2 + 0.05(-1)^2 + 0.15(3)^2 + 0.25(-2)^2 = 2.52$$

$$(c) \text{ Var}[Y] = \text{Var}[2X+1] = 4\text{Var}[X] = 4(E[X^2] - E[X]^2) = 4(2.52 - 0.5^2) = 9.08$$

$$(d) P(X \geq Y) = P(X = -1) + P(X = -2) = 0.3$$

14.

(a) There are $\binom{8}{4}$ ways to pick 4 lower level classes, and $\binom{10}{3}$ ways to choose 3 higher level classes, so there are

$\binom{8}{4} \binom{10}{3}$ valid curricula.

(b) We need to consider several different cases:

Suppose we do not choose L_1 . Then both L_2 and L_3 must be chosen; otherwise no higher level courses would be allowed. Thus, we need to choose 2 more lower level classes out of the remaining 5, and 3 higher level classes from the available 5.

We then obtain $\binom{5}{2} \binom{5}{3}$ valid curricula.

If we choose L_1 but choose neither L_2 nor L_3 , we have $\binom{5}{3} \binom{5}{3}$ choices.

If we choose L_1 and choose one of L_2 or L_3 , we have $2 \cdot \binom{5}{2} \binom{5}{3}$ choices. This is because there are two ways of choosing between L_2 and L_3 , $\binom{5}{2}$ ways of choosing 2 lower level classes from $[L_4, \dots, L_8]$ and $\binom{5}{3}$ ways of choosing 3 higher level classes from $[H_1, \dots, H_5]$.

Finally, if we choose L_1 , L_2 , and L_3 , we have $\binom{5}{1} \binom{10}{3}$ choices. The total is found by adding the above four.